| **Quantitative studies** | Ranabhat et al 2015 | Crawford et al, 2014 | Yadav et al, 2017 | Liu et al, 2015 | Adhikari et al, 2007 | Budhathoki et al, 2018 | Sharma et al, 2016 | Cardoso et al, 2018 | Amatya et al, 2018 | Padhye et al, 2003 | Shrestha et al, 2013 | Oster et al, 2011 | Parajuli et al, 2018 | Sapkota et al, 2013 | Parajuli et al, 2016 | Pramanik et al, 2010 | Paneru et al, 2012 | Katwal et al, 2016 | Rajbhandari et al , 2018 | Bauman et al, 2019 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1. Sampling methods | | | | | | | | | | | | | | | | | | | | |
| 1.1 Was the sample representative of the broader population? | 1 | N/A | 1 | 1 | 2 | 1 | 1 | 1 | 1 | 2 | 2 | 1 | 1 | 1 | 2 | 2 | 1 | 1 | 1 | 1 |
| 1.2 Was recruitment of participants appropriate to the study question? | 1 | N/A | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |
| 1.3 Adequate sample size (>100 or sample size calculation undertaken) | 1 | N/A | 1 | 2 | 2 | 1 | 1 | 1 | 1 | 1 | 3 | 1 | 1 | 2 | 1 | 1 | 1 | 1 | 1 | 1 |
| 1.4 Response rate reported and acceptable (≥70%) | 3 | N/A | 3 | 1 | 3 | 1 | 3 | 3 | 1 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 1 | 3 |
| 1.5 Control group is appropriate, clearly defined (if applicable) | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | 1 | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A |
| 2. Data collection | | | | | | | | | | | | | | | | | | | | |
| 2.1 Sample characteristics clearly described | 1 | N/A | 2 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |
| 2.2 Means of collecting data (e.g. assessment tool, questionnaire, etc) valid, reliable | 1 | N/A | 1 | 1 | 2 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 2 | 2 | 1 | 1 | 2 | 2 | 1 | 1 |
| 3. Data analysis / interpretation | | | | | | | | | | | | | | | | | | | | |
| 3.1 Potential confounders taken into account during the analysis and interpretation | 3 | N/A | 3 | 1 | 3 | 3 | 3 | 1 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 3 | 2 |
| 3.2 Tests for statistical significance undertaken, presented | 1 | N/A | 3 | 1 | 3 | 1 | 1 | 1 | 1 | 3 | 1 | 1 | 3 | 3 | 3 | 3 | 1 | 1 | 1 | 1 |
| | | | | | | | | | | | | | | | | | | | | |
| **Qualitative studies (adapted from RATS)** | | | | | | | | | | | | | | | | | | | | |
| 1. Study design | | | | | | | | | | | | | | | | | | | | |
| 1.1 Study design is appropriate to the research question | N/A | 1 | | N/A | N/A | 1 | N/A | N/A | 1 | N/A | | N/A | 1 | N/A | N/A | N/A | N/A | N/A | N/A | N/A |
| 1.1.2 Could a quantitative approach have worked better? | N/A | 1 | | N/A | N/A | 1 | N/A | N/A | 1 | N/A | | N/A | 1 | N/A | N/A | N/A | N/A | N/A | N/A | N/A |

| | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1.1.3 Justified why a particular method was chosen, e.g.: a)Interviews: experience, perceptions, behaviour, practice; b) Focus groups: group dynamics, convenience, non - sensitive topics; c) Ethnography: culture, organizational behaviour, interaction | N/A | 3 | | N/A | N/A | 1 | N/A | N/A | 1 | N/A | | N/A | 2 | N/A | N/A | N/A | N/A | N/A | N/A | N/A |
| 2. Sampling methods | | | | | | | | | | | | | | | | | | | | |
| 2.1 Criteria for selecting study sample is appropriate, e.g. purposive (diversity of opinion), random (generalizable to broader population), volunteer (hard to reach groups) | N/A | 1 | N/A | N/A | N/A | 1 | N/A | N/A | 1 | N/A | | N/A | 1 | N/A | N/A | N/A | N/A | N/A | N/A | N/A |
| 2.2 Details given of how recruitment was conducted and by whom | N/A | 1 | N/A | N/A | N/A | 3 | N/A | N/A | 1 | N/A | | N/A | 1 | N/A | N/A | N/A | N/A | N/A | N/A | N/A |
| 2.3 Details given on who chose not to participate and why | N/A | 3 | N/A | N/A | N/A | 3 | N/A | N/A | 1 | N/A | | N/A | 3 | N/A | N/A | N/A | N/A | N/A | N/A | N/A |
| 3. Data collection | | | | | | | | | | | | | | | | | | | | |
| 3.1 Collection of data is comprehensive and appropriate. E.g. a) Was the study setting appropriate? E.g. protection of confidentiality for sensitive discussions; b) Is the role of the researcher(s) appropriate? How might they bias the study and results? e.g. Do researchers occupy dual roles (clinician and researcher | N/A | 1 | N/A | N/A | N/A | 1 | N/A | N/A | 1 | N/A | | N/A | 3 | N/A | N/A | N/A | N/A | N/A | N/A | N/A |

| | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 4. Data analysis/interpretation | | | | | | | | | | | | | | | | | | | | |
| 4.1 Are interpretations clearly presented and supported adequately by evidence? | N/A | 1 | N/A | N/A | N/A | 1 | N/A | N/A | 1 | N/A | | N/A | 3 | N/A | N/A | N/A | N/A | N/A | N/A | N/A |
| 4.2 Indicators of quality | | | | | | | | | | | | | | | | | | | | |
| Descriptions of how themese were derived from the data (inductive or dedcutive) | N/A | 1 | N/A | N/A | N/A | 1 | N/A | N/A | 1 | N/A | | N/A | 3 | N/A | N/A | N/A | N/A | N/A | N/A | N/A |
| Semi quantification when appropriate | N/A | 3 | N/A | N/A | N/A | 3 | N/A | N/A | 3 | N/A | | N/A | 3 | N/A | N/A | N/A | N/A | N/A | N/A | N/A |
| Quote use appropriate, effective | N/A | 1 | N/A | N/A | N/A | 1 | N/A | N/A | 1 | N/A | | N/A | 3 | N/A | N/A | N/A | N/A | N/A | N/A | N/A |
| Analysis /presentation of negative/deviant cases, alternative explanations | N/A | 1 | N/A | N/A | N/A | 1 | N/A | N/A | 1 | N/A | | N/A | 3 | N/A | N/A | N/A | N/A | N/A | N/A | N/A |
| Method of reliability check (e.g. triangulation, independent review of data to contest themes) | N/A | 1 | N/A | N/A | N/A | 3 | N/A | N/A | 1 | N/A | | N/A | 3 | N/A | N/A | N/A | N/A | N/A | N/A | N/A |
| Are findings generalizable to a broader population? | N/A | 3 | N/A | N/A | N/A | 1 | N/A | N/A | 1 | N/A | | N/A | 2 | N/A | N/A | N/A | N/A | N/A | N/A | N/A |
| Total | 12 | 22 | 15 | 9 | 17 | 32 | 12 | 10 | 26 | 15 | 15 | 13 | 47 | 16 | 15 | 15 | 13 | 13 | 10 | 11 |
| Grading | * | ** | * | ** | + | + | + | ++ | ++ | + | - | ++ | - | + | + | + | + | + | ++ | ++ |
| *N/A marked under quantitative criteria for qualitative studies and vice versa. | | | | | | | | | | | | | | | | | | | | |
| | | | | | | | | | | | | | | | | | | | | |
| RUBRIC | | | | | | | | | | | | | | | | | | | | |
| ++ Low risk of bias: All or almost of the above criteria were fulfilled, and those that were not fulfilled were thought unlikely to alter the conclusions of the study. | | | | | | | | | | | | | | | | | | | | |
| + Medium risk of bias: Some of the above criteria were fulfilled, and those not fulfilled were thought unlikely to alter the conclusions of the study. | | | | | | | | | | | | | | | | | | | | |
| - High risk of bias: Few or no criteria were fulfilled, and the conclusions of the study were thought likely or very likely to alter with their inclusion. | | | | | | | | | | | | | | | | | | | | |